

Amygdala and emotions in animals



Klüver & Bucy syndrome (1937)

Excision of the mesial regions of both temporal lobes of monkey produced:

Hyperphagia

hypersexual behaviour

Psychic blindness (= insensitivity to the emotional value of stimuli) and excessive meekness

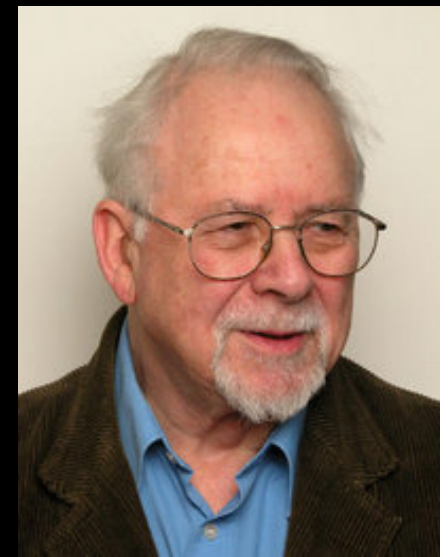
BEHAVIORAL CHANGES ASSOCIATED WITH ABLATION OF THE AMYGDALOID COMPLEX IN MONKEYS¹

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In a frequently cited study, Klüver and Bucy (7) noted striking behavioral changes following bilateral removal of the temporal lobes of the rhesus monkey. These changes were "visual agnosia," changes in dietary habits, increased tendency to examine objects ("hypermetamorphosis"), hypersexuality, and increased tameness. More recently, it has been demonstrated that the temporal lobe of the monkey can be functionally fractionated with respect to some of these general changes.

14). Understanding of the behavioral changes may not come until after experimental work isolates the critical relationships; contradictory results may be due, in part, to a lack of control of certain independent variables (such as daily feeding and handling procedures) which are uncovered only after such systematic work. Furthermore, many investigators have failed to report adequate histological reconstructions of lesions, a failing which is important in view of the relatively difficult surgical approach to



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50

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CHANGES IN VISUAL GNOSTIC FUNCTIONS AND EMOTIONAL BEHAVIOUR FOLLOWING UNILATERAL TEMPORAL POLE DAMAGE IN THE 'SPLIT-BRAIN' MONKEY

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SUMMARY

The performance of monkeys sustaining experimental lesions of the amygdala and medial temporal pole, or control lesions in the inferior temporal convexity or control sham operations, was studied grossly and with respect to the acquisition and extinction of conditioned avoidance and conditioned depression behavior. The following results were obtained:

1. In gross behavior, a marked increase in tameness and a weakening or disappearance of fear responses to previously aversive stimuli by amygdala animals.

2. A lower level for amygdala animals of avoidance and depression with respect to man as measured experimentally.

3. More rapid extinction by amygdala animals of conditioned avoidance and conditioned depression, when these behaviors had been established preoperatively.

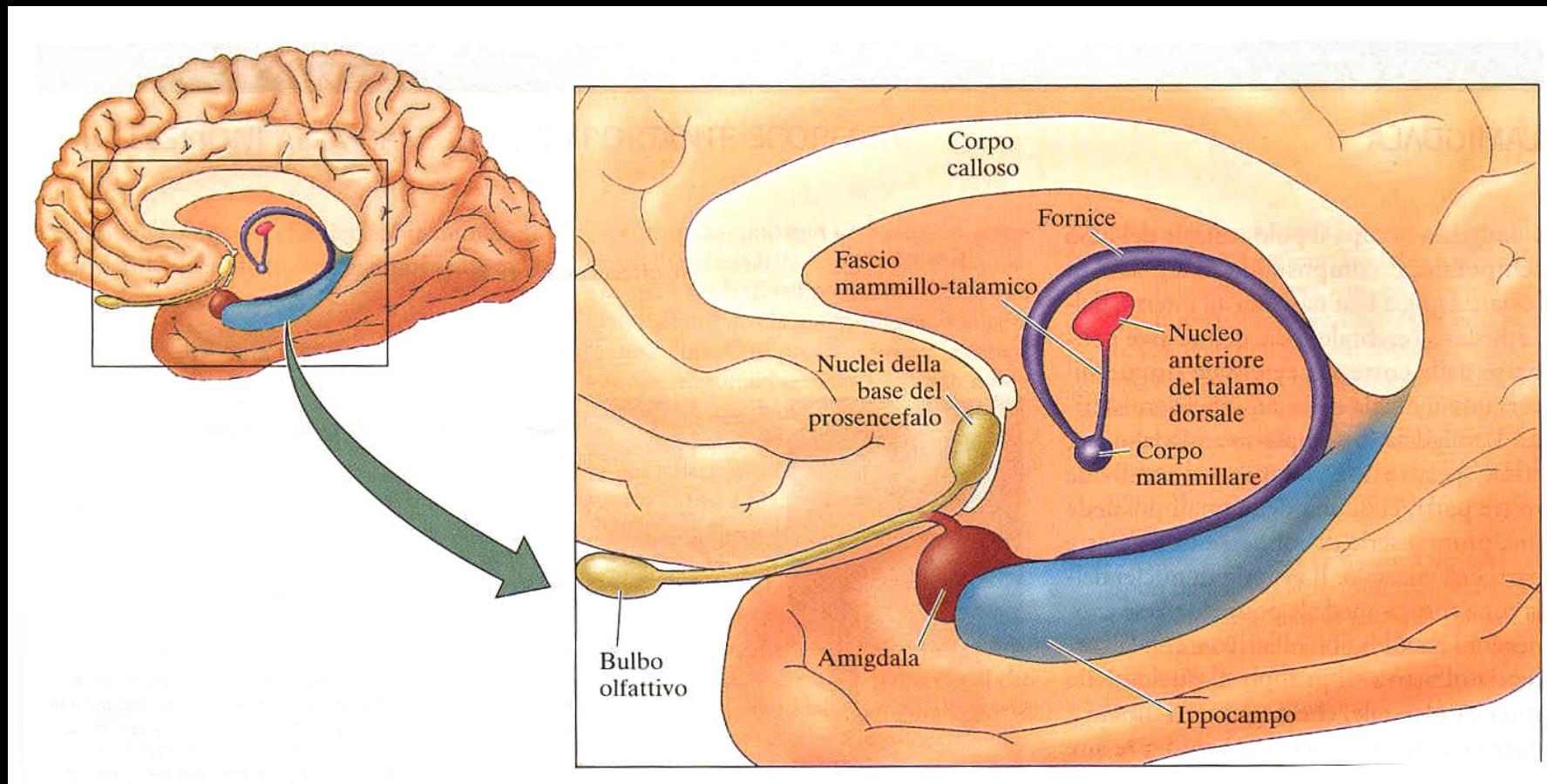
4. A slower rate of acquisition of these behaviors by amygdala animals when compared with sham animals, but not when compared with inferior temporal animals.

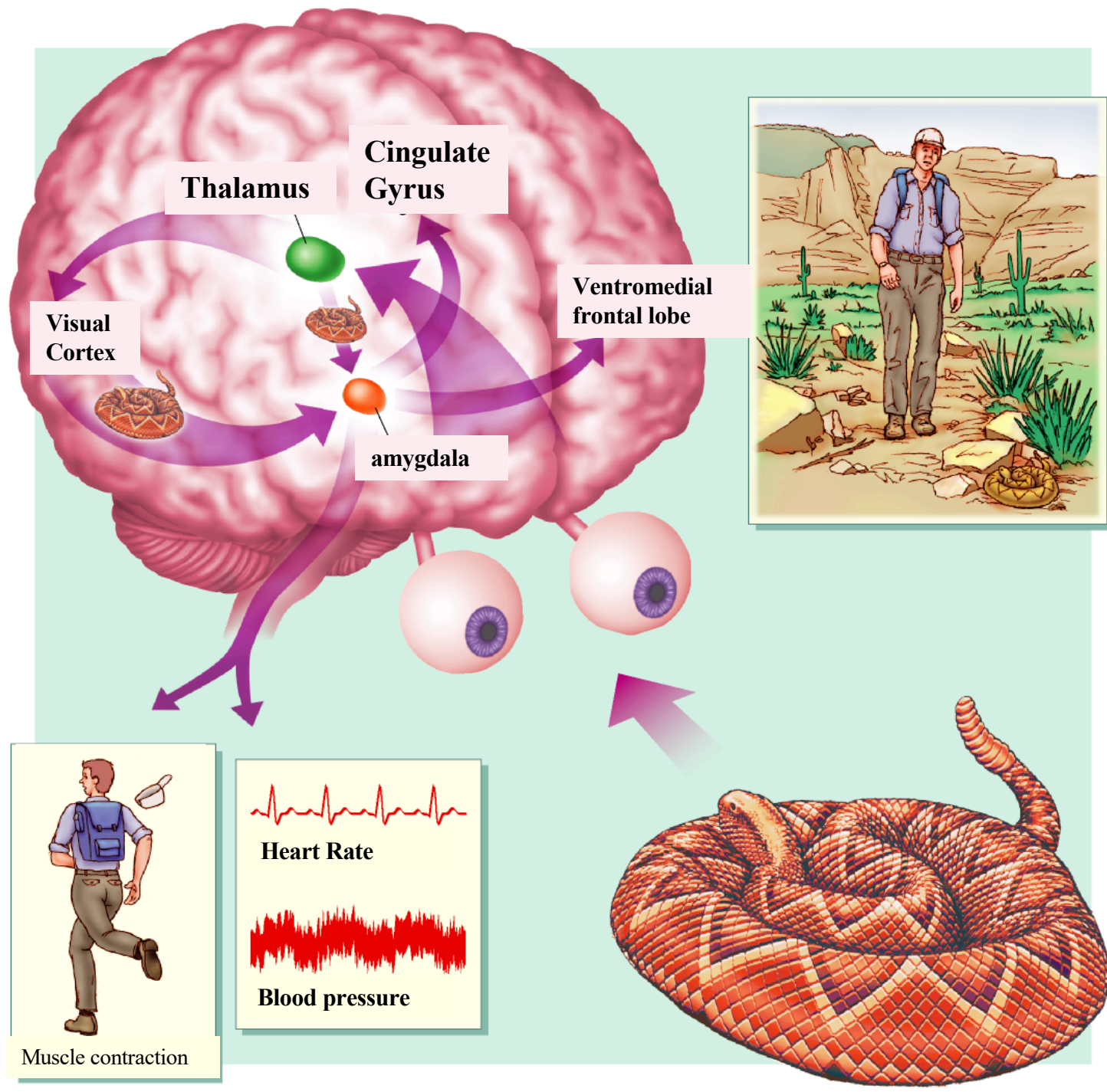
5. No differences among groups in retention of avoidance behavior.

Various possibilities that other behavioral changes may have secondarily produced the results are discussed. Evidence tends to rule out activity level changes, "decreased vigilance," or impairment in visual discrimination in amygdala animals. It is suggested that the effect of amygdalotomy is to make it difficult for animals to identify reinforcing stimuli.

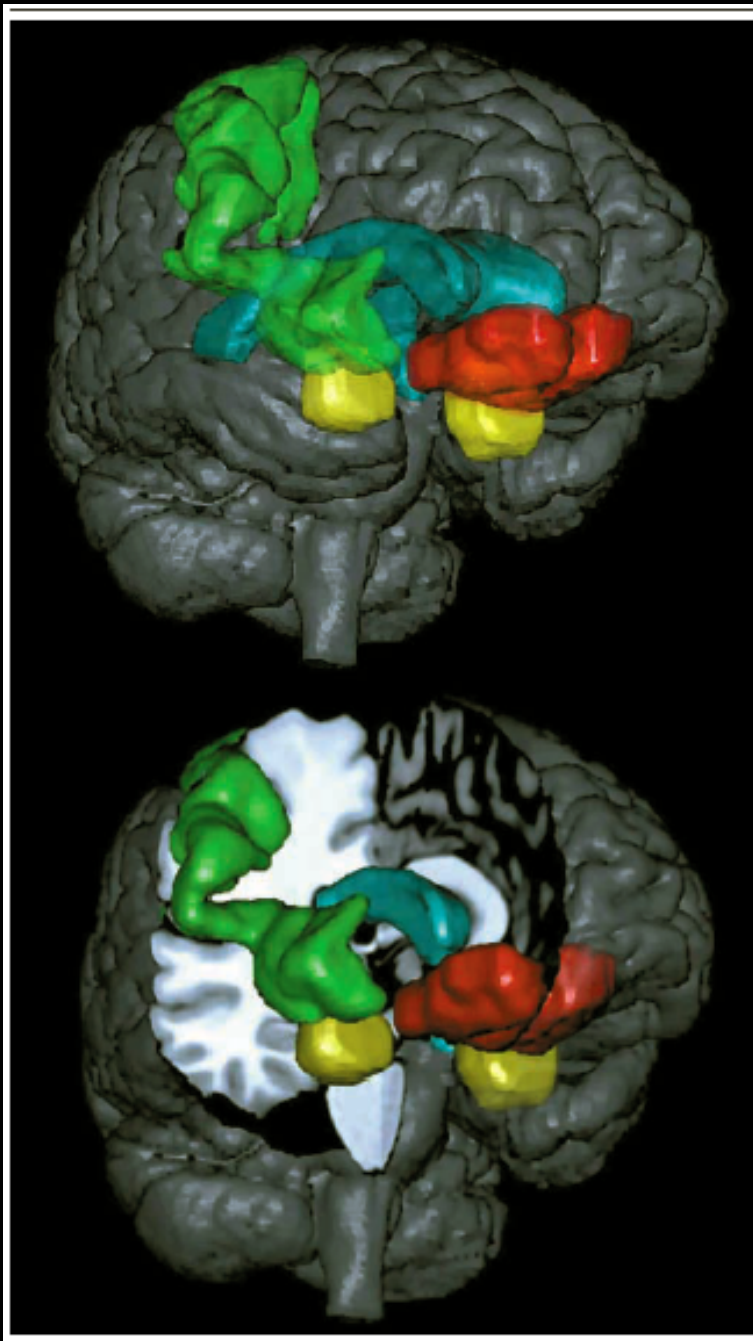
Weiskrantz 1956, J Comp Physiol Psychol

Limbic system and Triune Brain: MacLean (1949)





Structures for recognizing emotions



Amygdala



Ventro-medial prefrontal Cortex
(vmPFC)



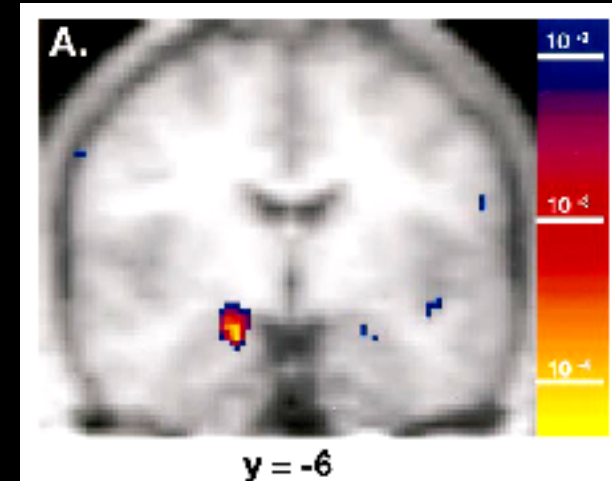
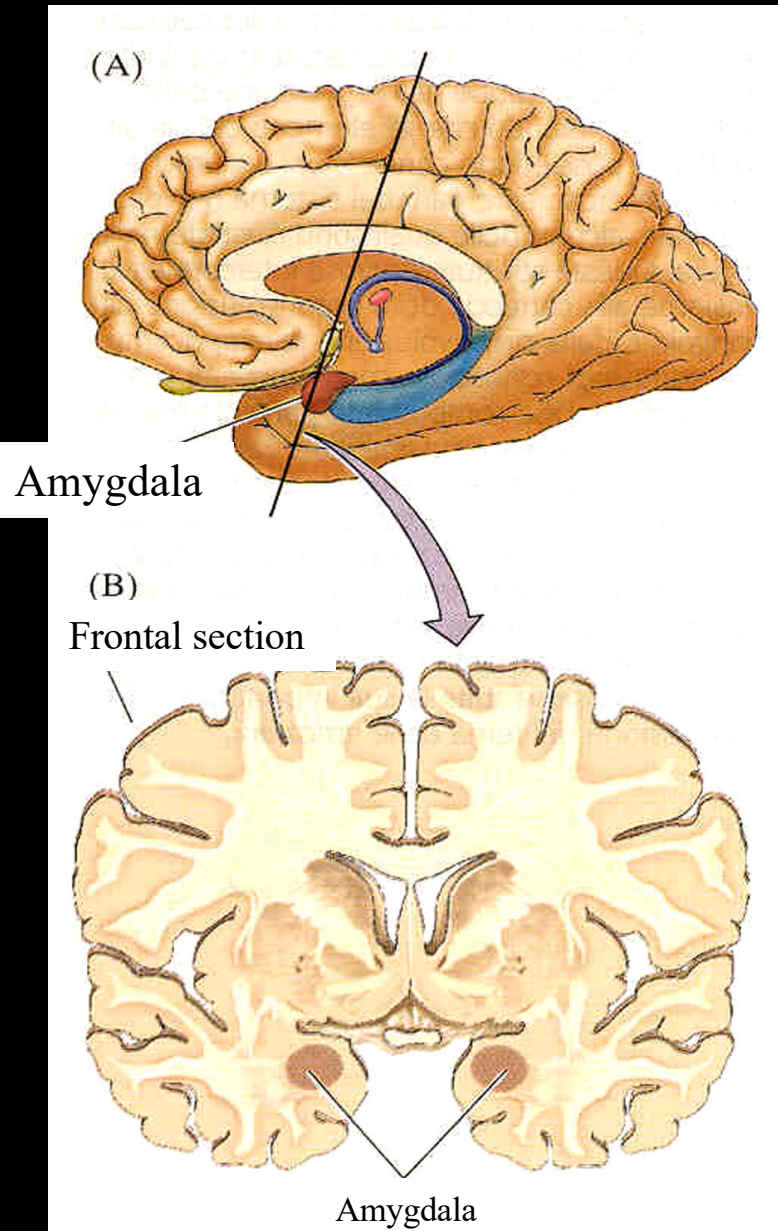
Somatosensory cortex (SI, SII)
and right insula



Lateral ventricles

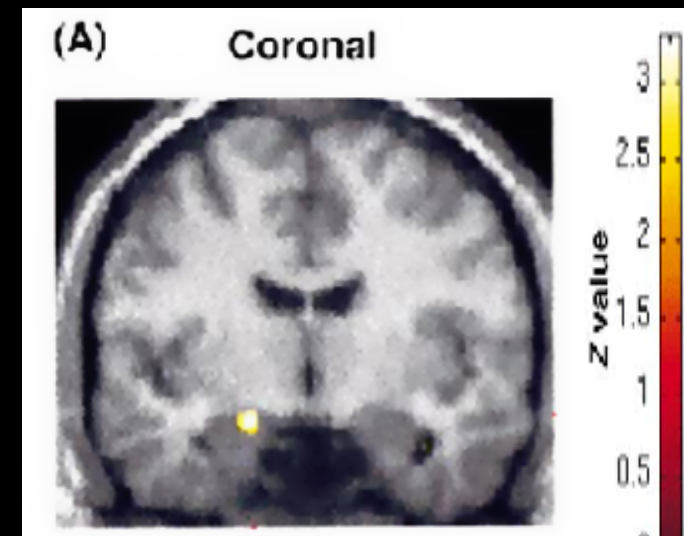
Amygdala (fMRI data in healthy subjects)

Fear



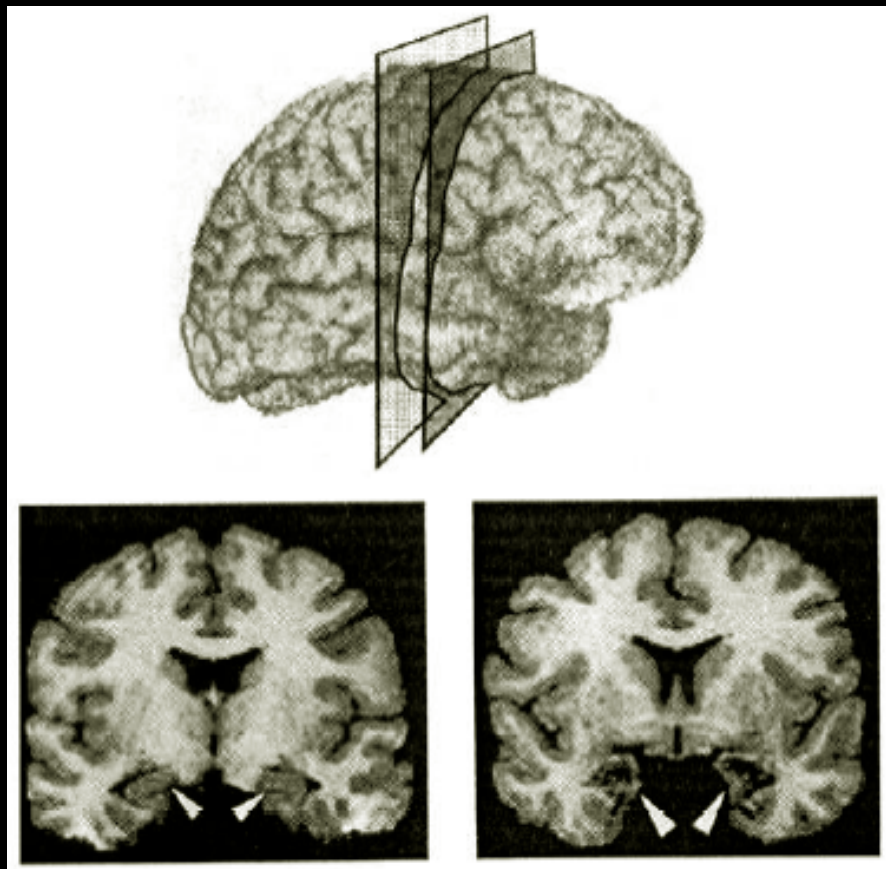
Whalen et al. *J. Neurosci.*, 18 (1), 411-418, 1998.

Sadness

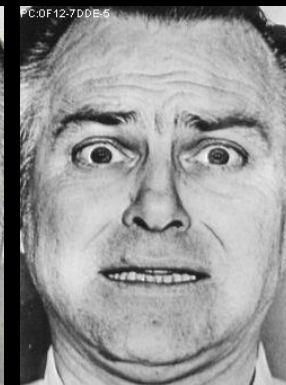
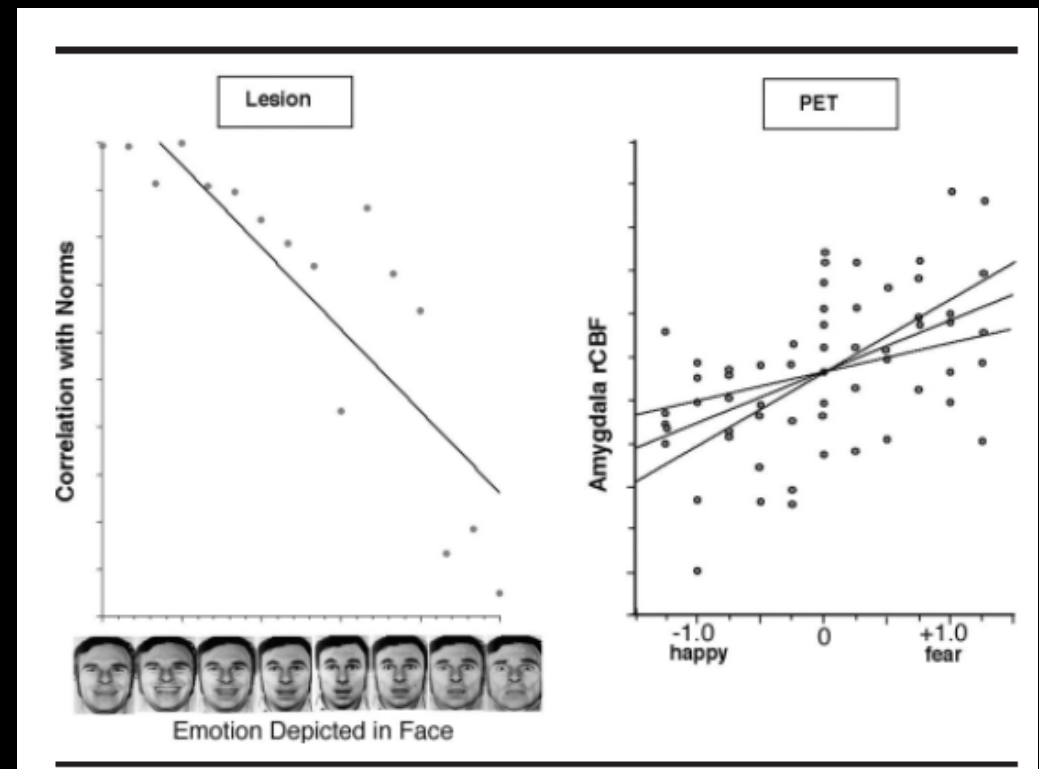


Whalen et al. *J. Neurosci.*, 18 (1), 411-418, 1998.

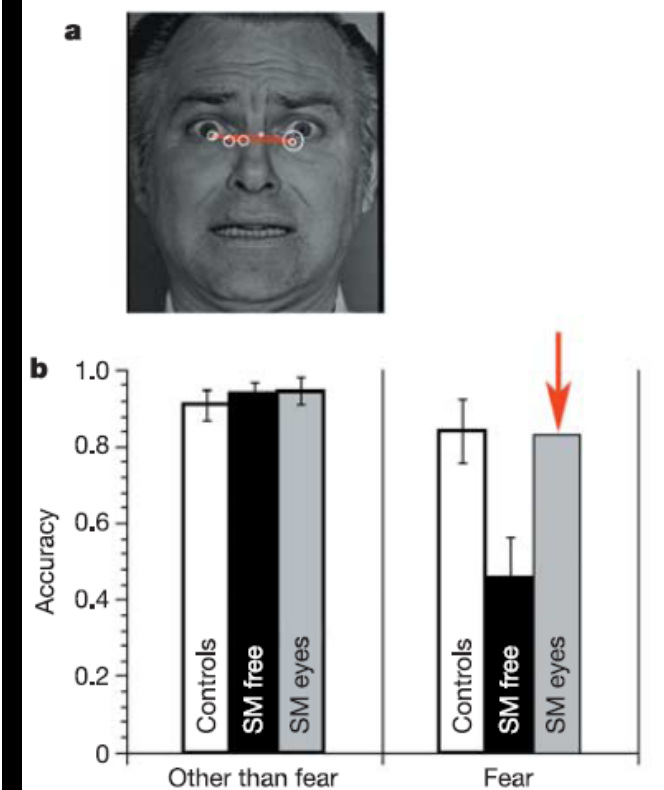
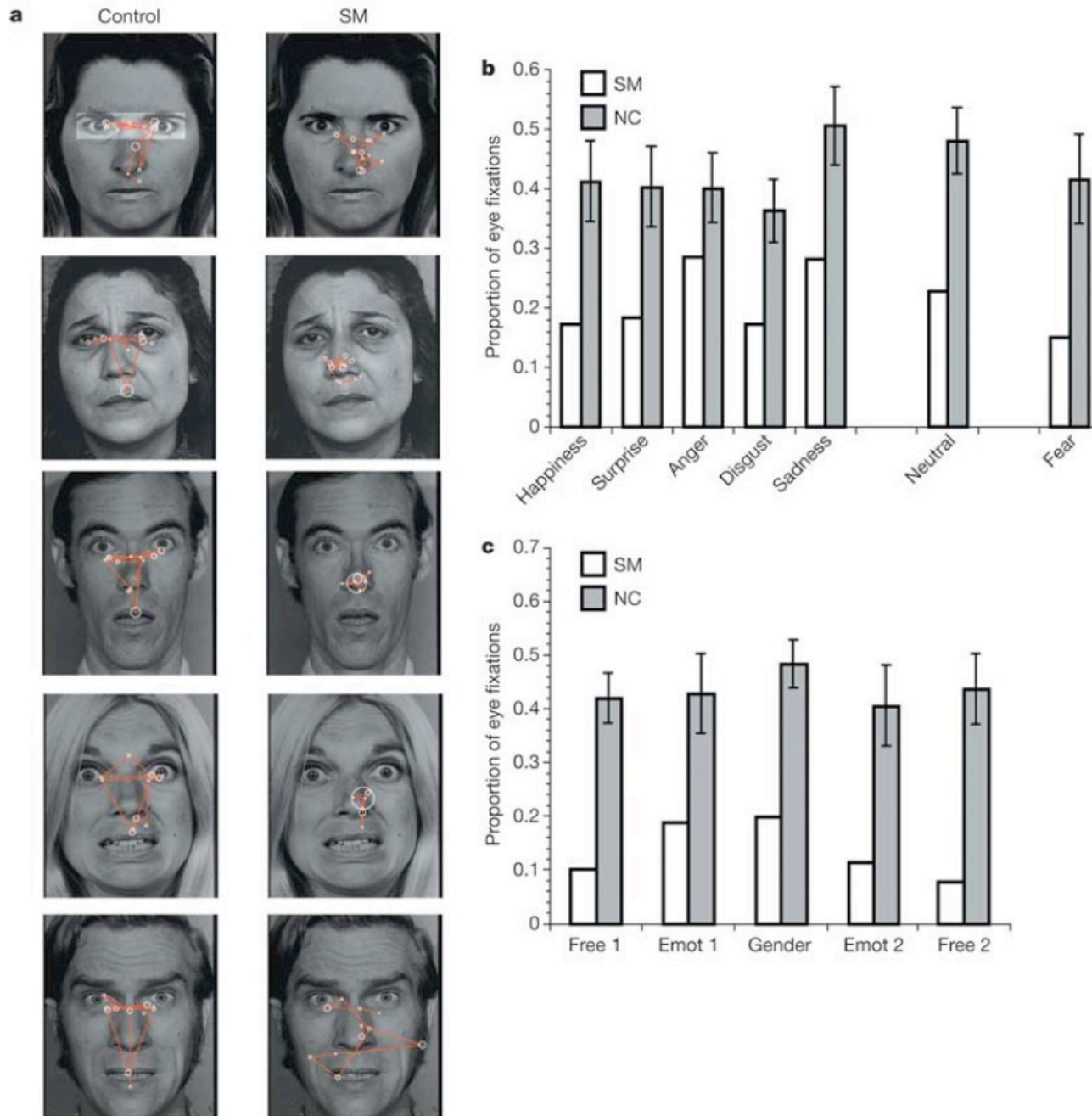
Amygdala (neuropsychological data in patients with focal lesions)



Adolphs et al.,
1994, Nature



Amygdala (neuropsychological data)



Adolphs et al.,
2005, Nature

Amygdala and social cognition

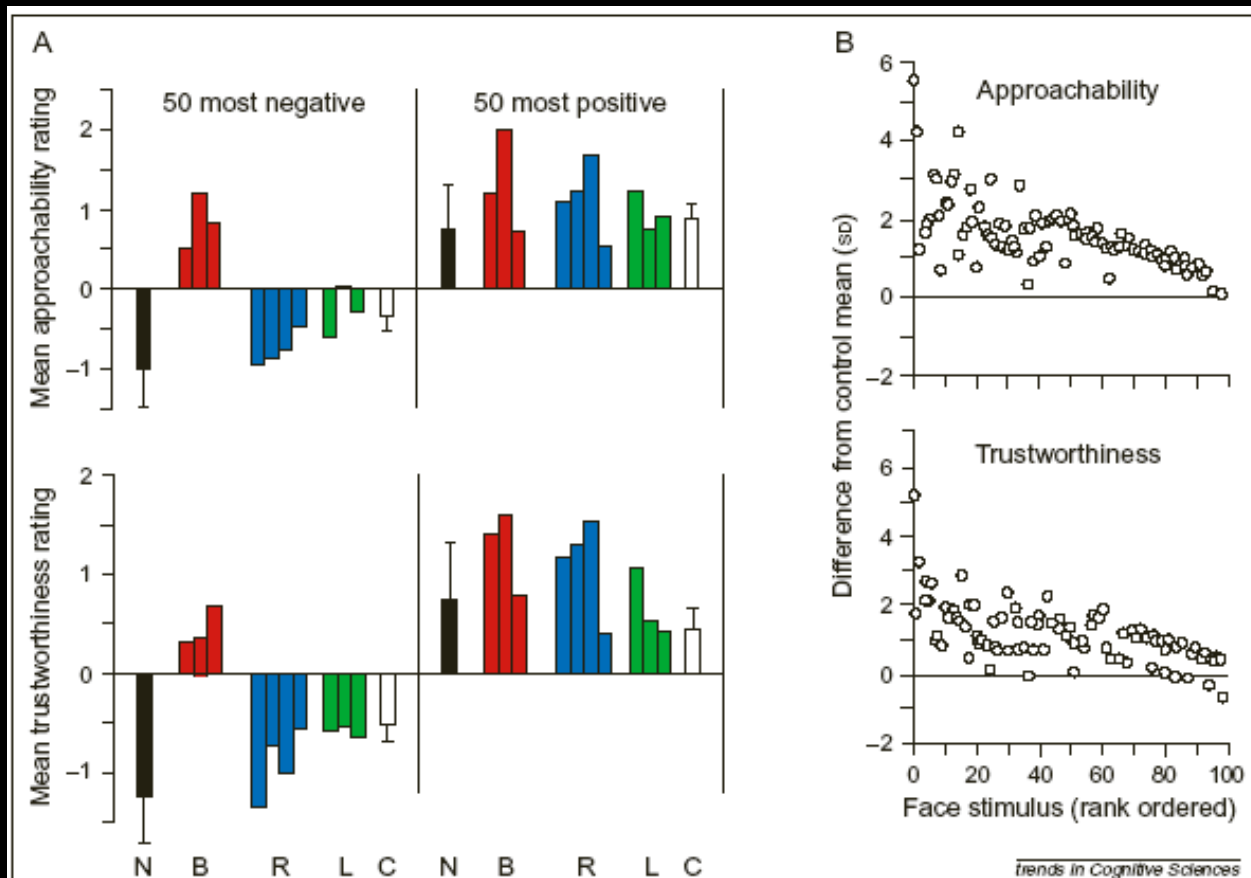


Fig. 2. Mean judgments of approachability and trustworthiness of 100 unfamiliar faces. (A) Means and SD for data from 46 normal controls are given at the left of each panel ('N'). Individual means are shown for each of three subjects with bilateral amygdala damage ('B'), four subjects with unilateral right ('R') and three with unilateral left ('L') amygdala damage. Group means and SEM are shown for seven brain-damaged controls with no damage to amygdala ('C'). The data, broken down into those obtained from the 50 faces that received the most negative mean ratings from normal controls (left) and the most positive (right) on each attribute, show that subjects with bilateral amygdala damage gave abnormally positive ratings to that half of the stimuli that normally receive the most negative ratings. (B) Standard deviations from the control mean are shown for one of the three subjects with bilateral amygdala damage (subject SM). Each face was judged on a scale of -3 (very unapproachable or untrustworthy) to +3 (very approachable or trustworthy). The data show that SM rated nearly all faces as positive. (Data redrawn from Ref. 43.)

Adolphs et al.,
1998, Nature